

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> First , we need to use the power formula to find the current . We know the power (40 watts) and the voltage (12 volts), so we can rearrange the formula $P = IV$ to find I , the current . $I = P / V$. $I = 40 / 12$. $I = 3 . 33$ amps . Now that we know the current , we can use Oh m 's law to find the resistance . $R = V / I$. $R = 12 / 3 . 33$. $R = 3 . 6$ oh ms . </think><answer> 3 . 33 amps , 3 . 6 oh ms

